

MAPPING THE MIND: VISUALIZING COGNITIVE BRAIN DYNAMICS WITH SPECTRAL ENTROPY TOPOMAPS IN R



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Abstract:

This project investigates spectral entropy as a spatially resolved indicator of cognitive activity in EEG data. Using the EEG Motor Movement/Imagery Dataset (EEGMMIDB) from PhysioNet, multichannel EEG recordings were extracted and processed. Entropy values were calculated across 64 electrodes after filtering and epoching around synthetic event markers to simulate cognitive task windows, with computations based on power spectral density estimates for each epoch and channel.

These entropy values were then spatially mapped to electrode positions following the 10–10 international system, creating topographic heatmaps that visualize entropy distribution across the scalp. This method is informed by neuroscience literature linking spectral entropy with mental workload and attention, providing a more interpretable alternative to opaque deep learning classifiers. Unlike traditional time-series EEG analysis, this approach offers a visual, data-driven summary of brain state complexity. The resulting topomaps effectively highlight regional entropy variations under different task types, emphasizing the value of open-access data, reproducible R code, and interpretable models in cognitive neuroscience. Future directions may include evaluating additional entropy indices or exploring clinical applications. This project exemplifies the value of combining statistics, neuroscience, and data visualization in a meaningful and accessible way.

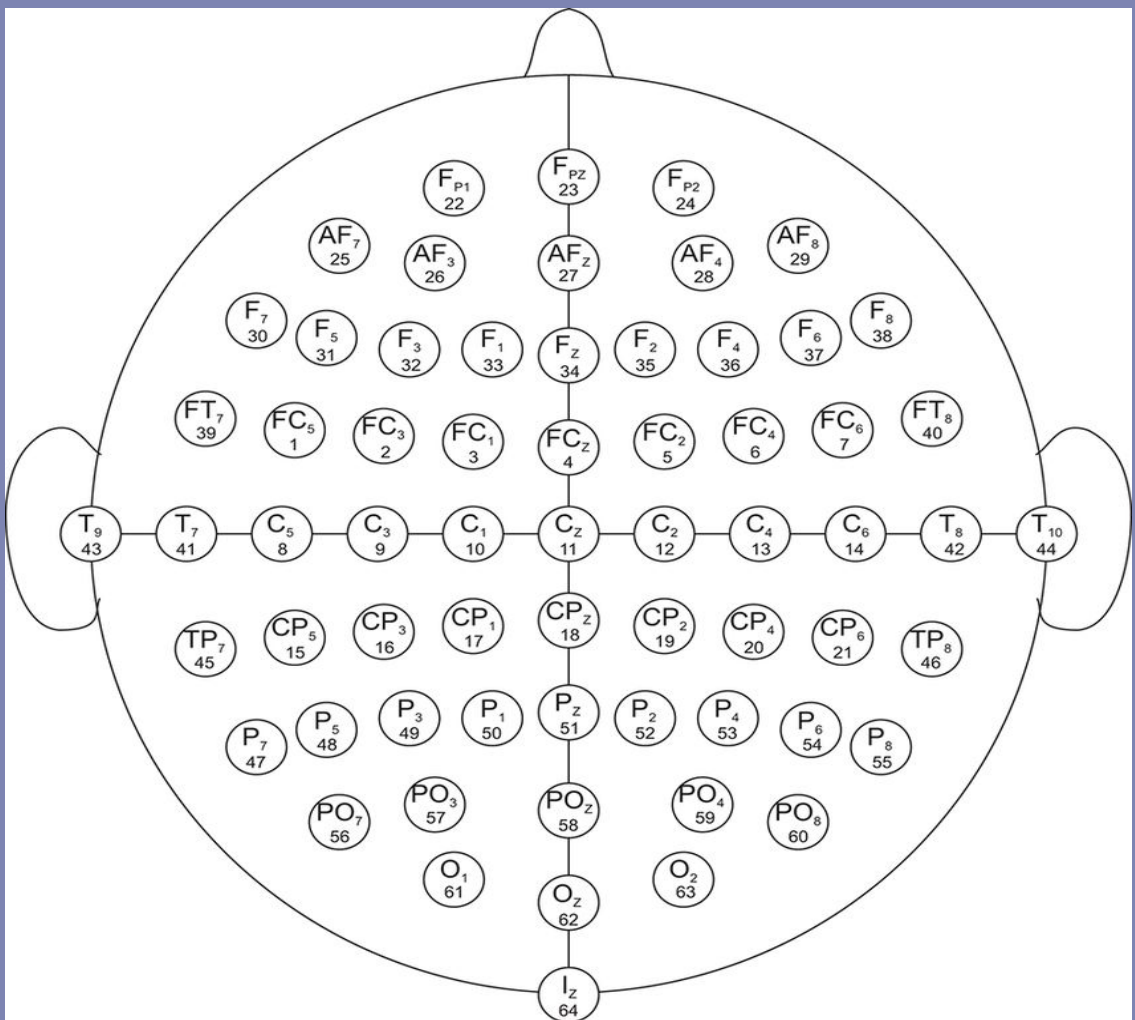


Figure 1. Conceptual figure showing the international, standard 10-10 system with 64 EEG electrodes placement. (Image included in the EEGMMIDB PhysioNet zip file)

Background Information:

- **EEG** (Electroencephalography) measures **electrical brain activity** through electrodes placed on the scalp, capturing dynamic neural oscillations in real-time.
 - These electrodes have **standardized placement systems** that use the nasion, inion, and left and right preauricular points to determine consistent positions across different individuals (**Figure 1**).
- **Spectral entropy** quantifies the **complexity or unpredictability** of EEG signals, making it a strong candidate for assessing cognitive load and mental states.
 - **Low spectral entropy** = regular, predictable signals
 - **High spectral entropy** = complex, varied signals
- Prior research has shown that spectral entropy **outperforms traditional EEG features** in distinguishing task difficulty and attention states (Zarjam et al., 2011; Tian et al., 2017).
- Deep learning models using topographic representations of EEG data (e.g., CNNs on EEG topomaps) have achieved high classification accuracy (Havugimana, 2021), but often **lack transparency or accessibility** for smaller-scale research.

Method/Materials:

- **Dataset:** PhysioNet EEGMMIDB
 - **Subject:** S001
 - **Sampling rate:** 160 Hz, 64 electrodes (10–10 layout)
 - **Runs:**
 - R01: Baseline (eyes open)
 - R02: Baseline (eyes closed)
 - R06: Task 4 (imagined movement of both fists or both feet)
- **Preprocessing:**
 - Bandpass filter: 0.5–45 Hz (remove slow drift/high-frequency noise)
 - Notch filter: 60 Hz (remove line noise)
 - Zero-phase Butterworth filters ([signal](#) package)
- **Event Simulation:** Dummy markers every 30s to simulate event timing
 - **Task labels:**
 - T0: Rest
 - T1: Onset of left/both fists movement
 - T2: Onset of right/both feet movement
- **Epoching Window:** 4-second epochs, starting 1s before each event
- **Entropy Computation:**
 - Power spectral density via [spectrum\(\)](#)
 - Normalized to probability distribution
 - Shannon entropy calculated per channel/epoch
 - Averaged by channel across all epochs per task
- **Visualization:**
 - Entropy values placed on 10–10 scalp layout
 - Interpolated topomaps using [interp](#) + [ggplot2](#)
 - “Spectral” palette (red = high entropy, blue = low)
 - Scalp outline and electrodes for spatial context

Results:

Three EEG spectral entropy topomaps were generated using a shared color scale (5.0–7.5) to compare different conditions. Notable spatial differences in entropy were observed across 64 scalp electrodes:

- **Baseline, Eyes Closed (Figure 2):**
 - Entropy was highest in anterior and superior regions (prefrontal and central electrodes).
 - This suggests rich, spontaneous activity during the resting state.
 - Posterior areas showed lower entropy.
- **Baseline, Eyes Open (Figure 3):**
 - Entropy was more uniformly distributed.
 - There was a mild decrease in frontal regions.
 - Slightly elevated entropy was observed in posterior-occipital channels.
- **Task 4, Imagined Bilateral Movement (Figure 4):**
 - Entropy dropped significantly in fronto-central regions.
 - A distinct low-entropy “valley” appeared near the motor cortex.
 - This shift suggests structured, lower-complexity neural patterns during motor imagery

EEG Spectral Entropy Topomap (Baseline, Eyes Closed)

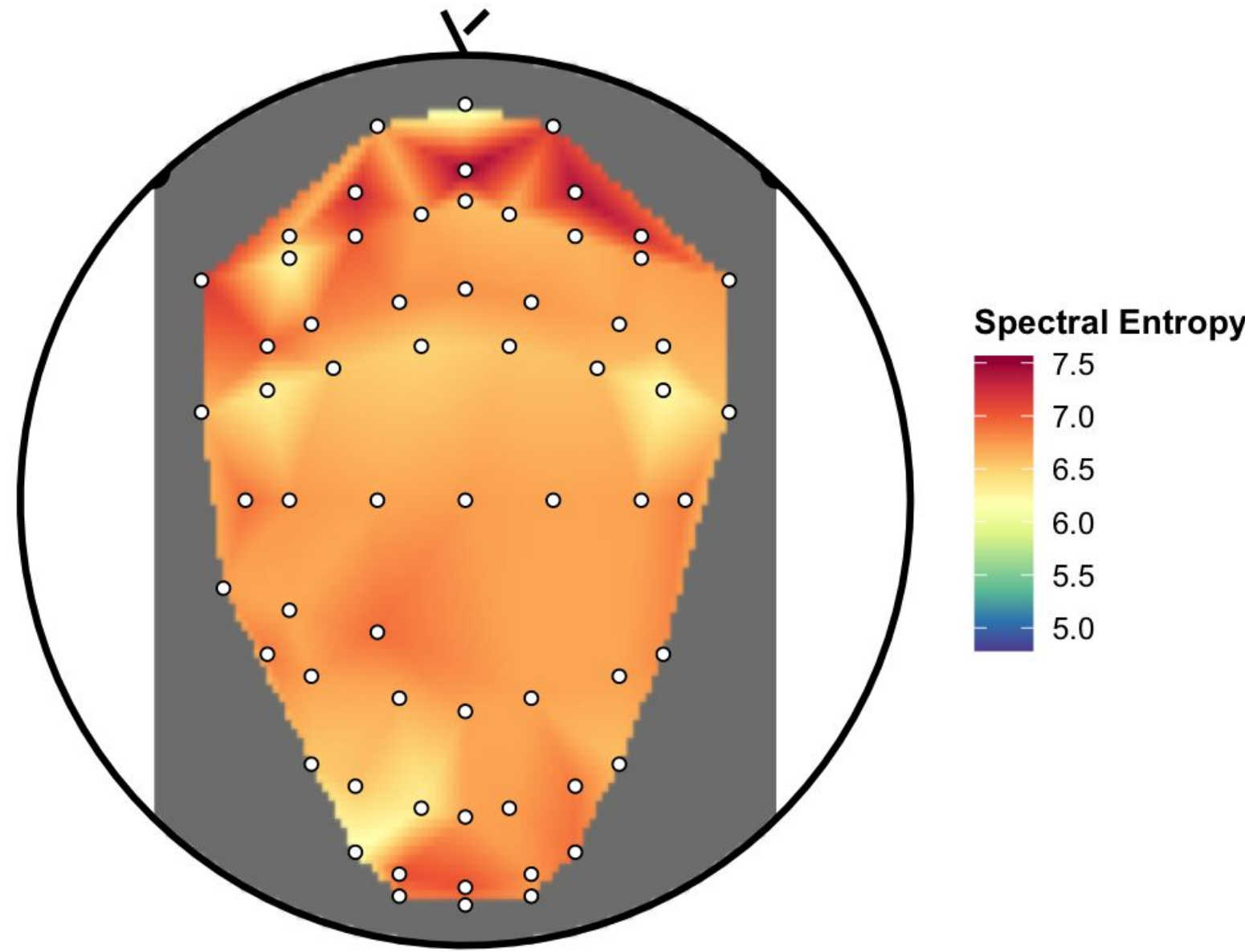


Figure 2. Average spectral entropy across EEG channels during the resting-state, eyes-closed condition. Entropy is highest in anterior and central regions, reflecting elevated spontaneous neural complexity during internal thought and reduced sensory input.

EEG Spectral Entropy Topomap (Baseline, Eyes Open)

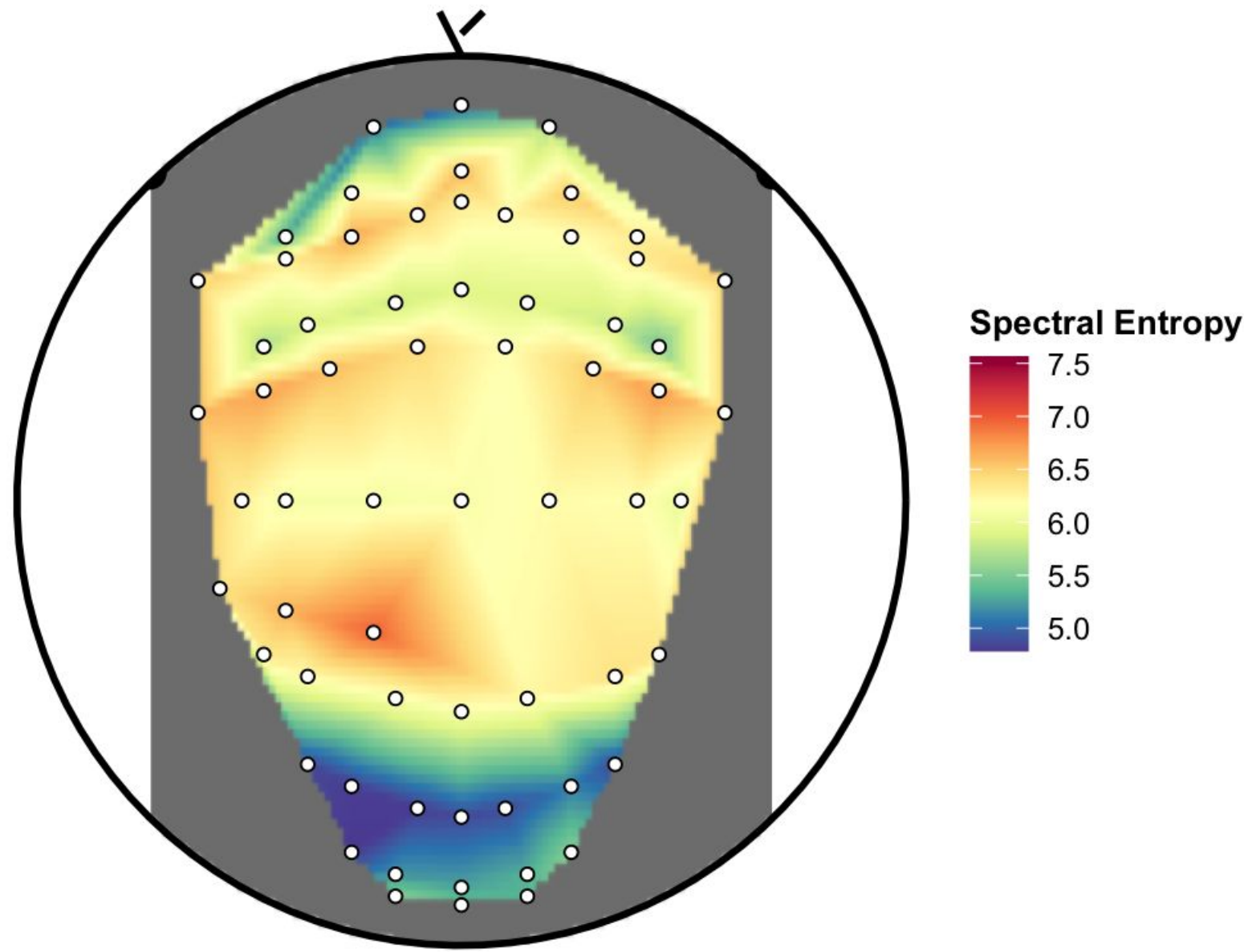


Figure 3. Spectral entropy distribution while Subject S001 rests with their eyes open. Compared to the eyes-closed state, frontal entropy is reduced, and posterior regions show relatively increased entropy, likely reflecting visual processing and alertness.

EEG Spectral Entropy Topomap (Task 4)

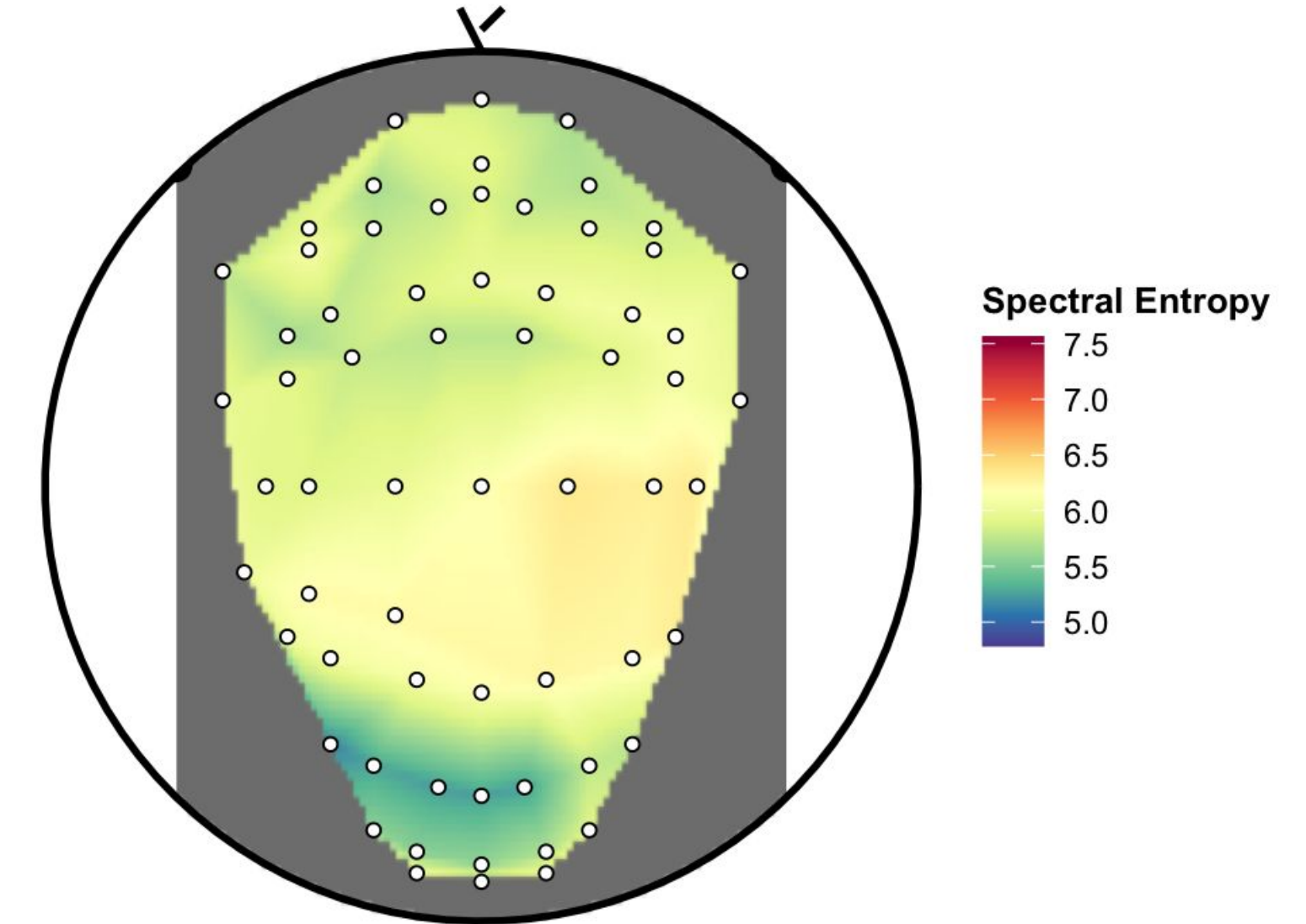


Figure 4. Entropy pattern during imagined bilateral fist or foot movement. A pronounced decrease in entropy is visible over fronto-central regions, consistent with engagement of sensorimotor areas and reduced signal complexity during focused motor imagery.

Discussion/Conclusion:

- Spectral entropy **consistently decreased during task engagement**, especially within sensorimotor networks. This suggests more organized neural firing during tasks.
 - A clear reduction in entropy over **sensorimotor areas** was observed during imagined movement, likely linked to motor planning in the **precentral gyrus**.
- The eyes-closed condition exhibited elevated entropy, consistent with **increased alpha activity** and **default mode network engagement**.
- Using a consistent color scale (5.0–7.5) across all topomaps was essential for accurate and meaningful comparisons, preventing misinterpretation of differences.
- This project demonstrates that entropy-based EEG visualizations effectively **localize** and **differentiate brain states**.
 - Spectral entropy distinctly **decreases during imagined motor activity** (fronto-central channels), and shared visual scales enhance clarity.
- Overall, EEG entropy topomaps are an effective and transparent tool for exploring **cognitive load, motor imagery, and brain state complexity**.

Future Work:

- **Additional Entropy Indices:** Looking beyond spectral entropy, the implementation of more entropy indices into the topomap visualization could provide more complex and varied insights into cognitive load.
- **Real-time Feedback Systems:** This approach could be integrated into systems that provide immediate feedback based on brain state, potentially for objectively classifying and monitoring cognitive or mental workload in various settings.
- **Clinical Population Comparisons:** The methodology can be extended to compare brain states across different clinical populations, such as in studies of cognitive decline (e.g., Alzheimer's disease) or neurorehabilitation.

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